

11-matching-flow: 匹配与网络流 (Matching and Network Flow)

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3 Theorems + 1 Algorithm

To **maximize** the size of a mathematical structure $\mathcal{S}$ in G



Theorem

$\mathcal{S}$ is maximum *iff* G does not contain $\mathcal{S}$ -augmenting objects.

Algorithm

Repeatedly finding $\mathcal{S}$ -augmenting objects until no more ones exist.

To **maximize** the size of a mathematical structure $\mathcal{S}$ in G



To **minimize** the size of its **dual** mathematical structure $\mathcal{S}'$ in G

Theorem (Weak Duality Theorem)

The size of a maximum $\mathcal{S} \leq$ The size of a minimum $\mathcal{S}'$

Theorem (Strong Duality Theorem)

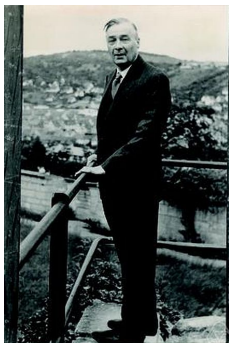
The size of a maximum $\mathcal{S} =$ The size of a minimum $\mathcal{S}'$

let's get
married
today

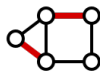
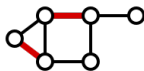
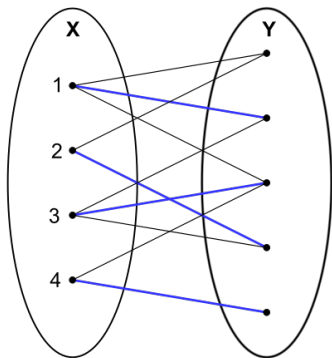


The Marriage Problem (Philip Hall, 1935)

If there is a finite set of **girls**, each of whom knows several **boys**,
under what conditions can all the girls marry boys in such a way that
each girl marries **a distinct boy** that she knows?



Philip Hall (1904 ~ 1982)



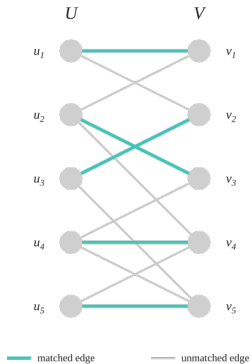
Maximal Matching



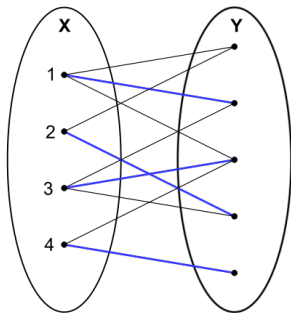
Maximum Matching

Definition (Matching (匹配))

A **matching** in a graph G is a set of edges with **no shared endpoints**.



Perfect Matching



X -Perfect Matching

Definition (X -Perfect Matching (X -Saturating Matching))

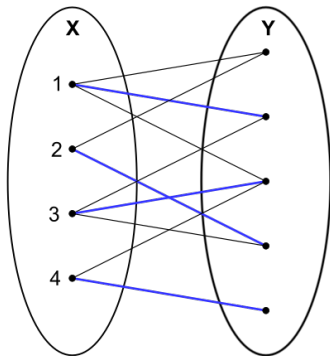
Let $G = (X, Y, E)$ be a bipartite graph.

An X -perfect matching of G is a matching which covers each vertex in X .

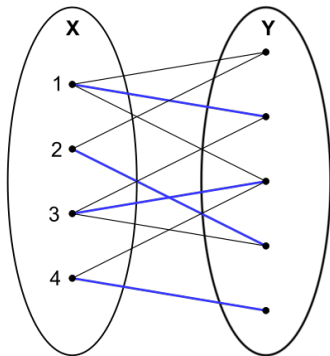
The Marriage Problem (Philip Hall, 1935)

Let $G = (X, Y, E)$ be a bipartite graph.

Is there an **X-perfect matching** of G ?



$$|X| \leq |Y|$$



$$\forall W \subseteq X. |W| \leq |N(W)|$$

Theorem (Hall Theorem; 1935)

Let $G = (X, Y, E)$ be a bipartite graph.

There is an *X -perfect matching* of G if and only if

$$\forall W \subseteq X \quad |W| \leq |N_G(W)|.$$

By induction on the number $|X|$ of vertices in X .

Basis Step: $|X| = 1$. $|X| \leq |N_G(X)|$. I am married!

Induction Hypothesis: Suppose that it holds if $|X| < m$.

Induction Step: Consider the case $|X| = m$.

How to reduce the case $|X| = m$ to a case $|X'| < m$?

Consider the case $|X| = m$.

► CASE I: Every $k < m$ girls in X know $\geq k + 1$ boys in Y .

Take any girl x and marry her to any boy y she knows.

$$G' = G - \{x, y\}$$

Hall's Condition still holds for G' .

$$\forall W \subseteq X - \{x\}. |W| \leq |N_{G'}(W)|$$

There is an $(X - \{x\})$ -perfect matching in G' .

Therefore, there is an X -perfect matching in G .

- CASE II: There is a set of $k < m$ girls in X who know exactly k boys in Y .

There k girls can be married **by induction** to the k boys.

Hall's Condition holds for these k girls and the k boys.

$$G' = G - \{\text{these } k \text{ girls}\} - \{\text{these } k \text{ boys}\}$$

Hall's Condition still holds for G' .

By contradiction:

Suppose that in G' , there are $l \leq m - k$ girls who know $< l$ boys.

Then, in G , there are $k + l$ girls who know $< k + l$ boys.

There is an $(X - \{\text{these } k \text{ girls}\})$ -perfect matching in G' .

Therefore, there is an X -perfect matching in G .

Theorem (Hall Theorem; 1935)

Let $G = (X, Y, E)$ be a bipartite graph.

$$\forall W \subseteq X \quad |W| \leq |N_G(W)|.$$

Definition (M -alternating Paths)

Let M be a matching. An M -alternating path is a path that alternates between edges in M and edges not in M .



Definition (M -augmenting Paths)

An M -augmenting path is an M -alternating path whose endpoints are unsaturated by M .

Theorem (Hall Theorem; 1935)

Let $G = (X, Y, E)$ be a bipartite graph.

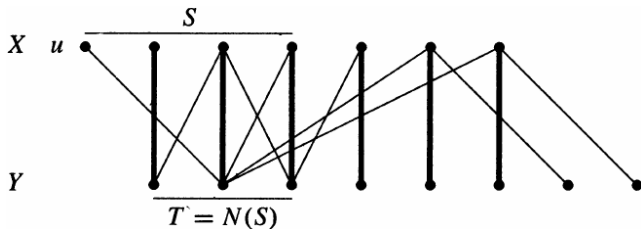
There is a **X** -perfect matching of G if and only if

$$\forall W \subseteq X \quad |W| \leq |N_G(W)|.$$

By contradiction.

Suppose that there is *no* X -perfect matching in G .

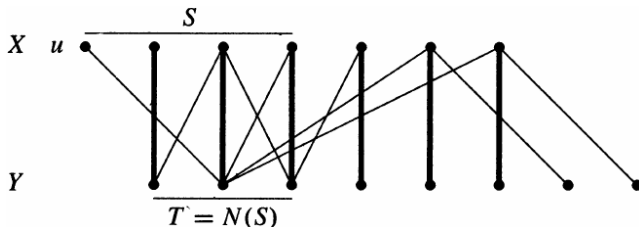
Let M be a *maximum* matching in G .



We show that Hall's Condition is violated for *some* $S \subseteq X$.

We will construct $S \subseteq X$ based on M .

Let $u \in X$ be a vertex of X *not* saturated by M .



Consider all *M -alternating paths* starting from u .

$T \triangleq$ the set of vertices in Y reachable from u by M -alternating paths.

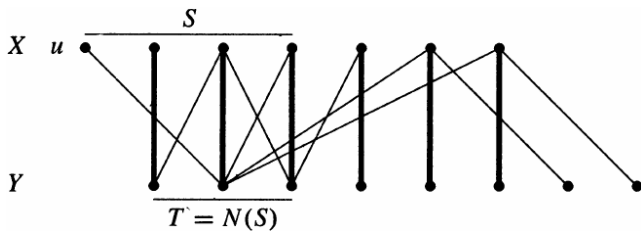
$S \triangleq$ the set of vertices in X reachable from u by M -alternating paths.

We will show that

$$T = N(S) \wedge |T| = |S - \{u\}|$$

$$|N(S)| = |T| = |S| - 1 < |S|$$

$$|T| = |S - \{u\}|$$



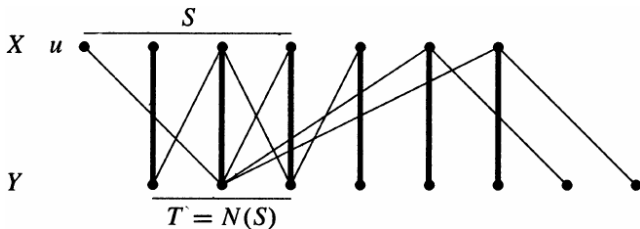
We show that there is a **bijection** from T to $S - \{u\}$.

M matches T with $S - \{u\}$.

Every vertex in T goes via an edge in **M** to a vertex in $S - \{u\}$.

Every vertex in $S - \{u\}$ is reached by an edge in **M** from a vertex in T .

$$T = N(S)$$



M matches T with $S - \{u\} \implies T \subseteq N(S)$

We need to show that $N(S) \subseteq T$

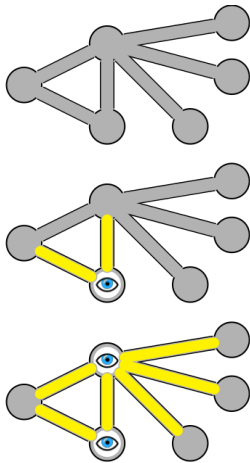
Suppose **by contradiction** that $v \in S$ has a neighbor $y \in Y - T$.

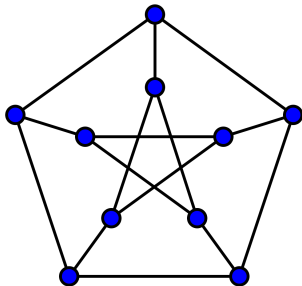
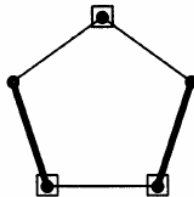
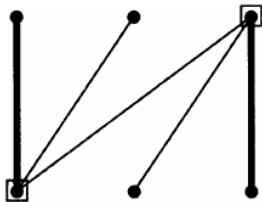
It must be $vy \notin M$.

There exists an M -alternating path to y from u . $\implies y \in T$.

Definitions (Vertex Cover (点覆盖))

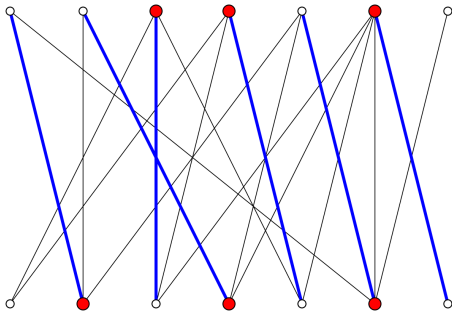
A **vertex cover** of a graph G is a set $Q \subseteq V(G)$ that **covers** all edges.





Theorem (Weak Duality Theorem (弱对偶定理))

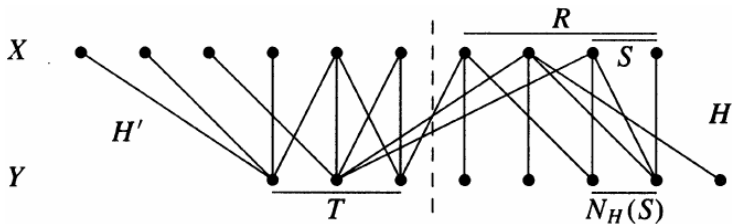
Let G be a graph. The maximum size of a matching in G
 $\leq$ the minimum size of a vertex cover of G .



Theorem (König (1931), Egerváry (1931))

Let G be a *bipartite* graph. The maximum size of a matching in G
 $=$ the minimum size of a vertex cover of G

Given a minimum vertex cover Q ,
we construct a matching M of the same size.



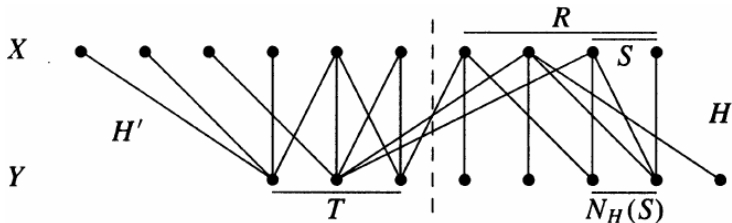
$$R = Q \cap X \quad T = Q \cap Y$$

$H \triangleq (R \cup (Y - T))$ -induced subgraph of G

$H' \triangleq (T \cup (X - R))$ -induced subgraph of G

H has a R -perfect matching and H' has a T -perfect matching.

H has a R -perfect matching and H' has a T -perfect matching.



To show that Hall's Condition holds for H (and R).

By contradiction.

$$\exists S \subseteq R. |N_H(S)| < |S|$$

$T \cup (R - S + N_H(S))$ is a smaller vertex cover than Q .

Augmenting Path Algorithm for Maximum Bipartite Matching

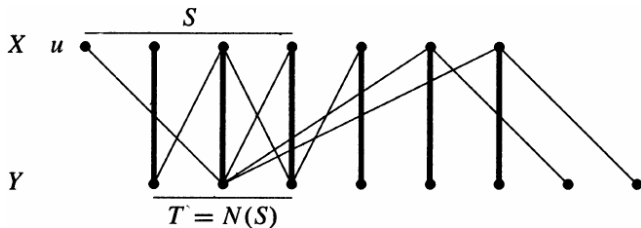
- ▶ $M = \emptyset$
- ▶ **While** there **is** an M -augmenting path P :
 - ▶ Enlarge M by P : $M \leftarrow M \oplus P$
- ▶ **Return** M

How to **find M -augmenting paths** efficiently?

How to prove that **the M returned** is a **maximum** matching?

Augmenting Path Algorithm for Maximum Bipartite Matching

- ▶ $M = \emptyset$
- ▶ **While** there **is** an M -augmenting path P :
 - ▶ Enlarge M by P : $M \leftarrow M \oplus P$
- ▶ **Return** M

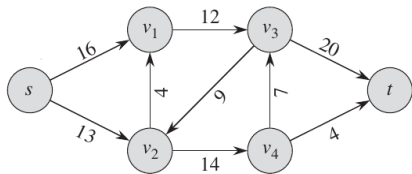
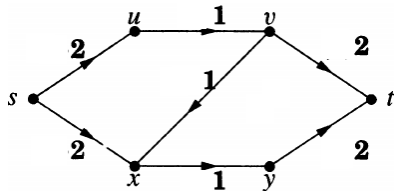


When the algorithm terminates,
it actually finds a **vertex cover** $T \cup (X - S)$ of size M .

Definition (Network (网络))

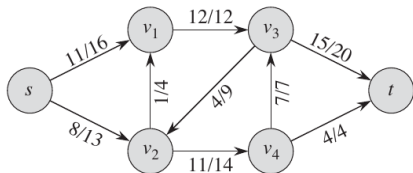
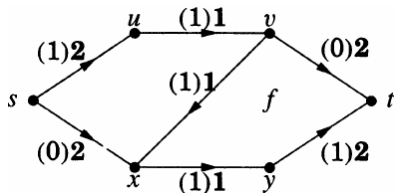
A **network** is a **digraph** with

- ▶ a distinguished **source vertex** s ,
- ▶ a distinguished **sink vertex** t ,
- ▶ a **capacity** $c(e) \geq 0$ on each edge e



Definition (Flow (流))

A **flow** f is a **function** that assigns a value $f(e)$ to each edge e .



Definition (Feasible Flow (可行流))

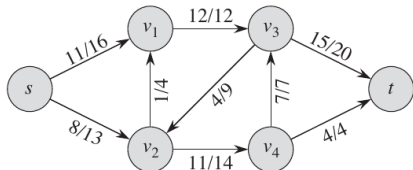
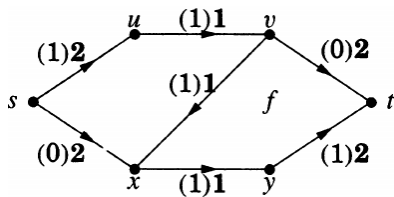
A flow f is **feasible** if it satisfies

Capacity Constraints:

$$\forall e \in E \quad 0 \leq f(e) \leq c(e)$$

Flow Conservation:

$$\forall v \in V \setminus \{s, t\} \quad f^+(v) = f^-(v)$$



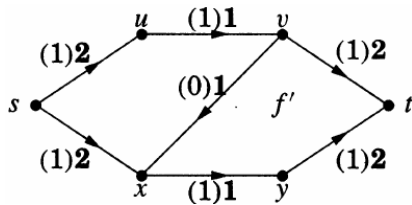
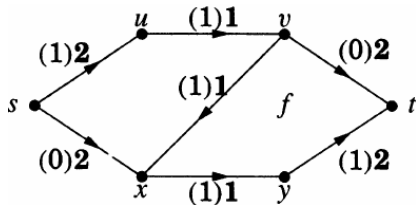
$$f^+(v) = \sum_{(v,w) \in E} f(v,w)$$

$$f^-(v) = \sum_{(u,v) \in E} f(u,v)$$

Definition (Value (值))

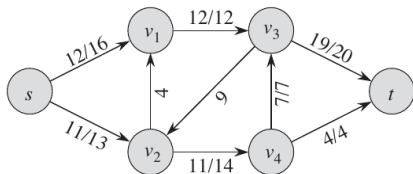
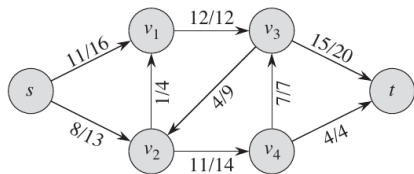
The **value** $\text{val}(f)$ of a **flow** f is

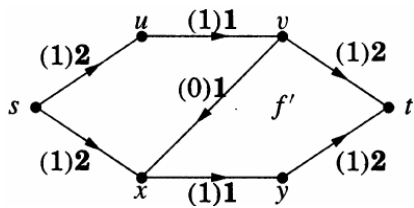
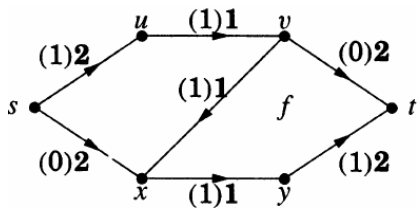
$$\text{val}(f) = f^-(t) = f^+(s).$$



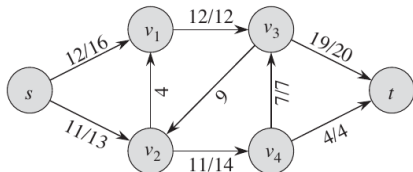
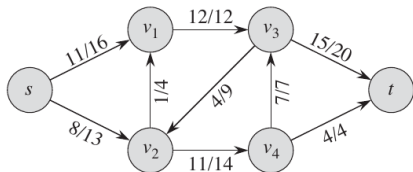
Definition (Maximum Flow (最大流))

A **maximum flow** is a **feasible flow** of maximum **value**.





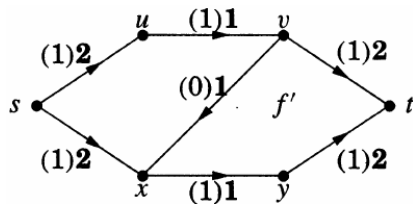
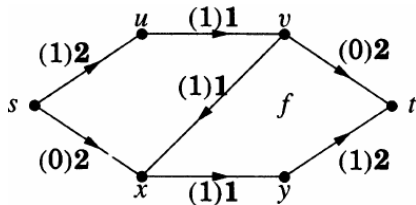
$s \sim t$ path : $s - x - v - t$



$s \sim t$ path : $s - v_1 - v_2 - v_3 - t$

$s \sim t$ path : $s - v_2 - v_3 - t$

$$s \sim t \text{ path} : s - x - v - t$$



Definition (f -augmenting Paths (增广路径))

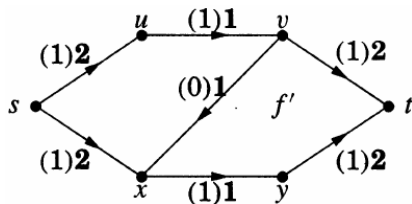
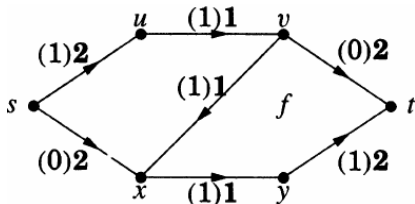
When f is a feasible flow, an **f -augmenting path** is a $s \sim t$ path P in the underlying graph such that for each edge $e \in E(P)$,

- (a) if P follows e in the forward direction, then $f(e) < c(e)$;
- (b) if P follows e in the backward direction, then $f(e) > 0$.

Definition (f -augmenting Paths)

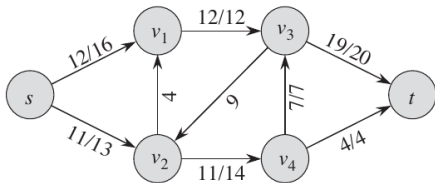
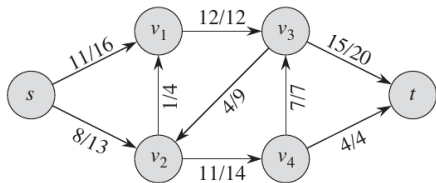
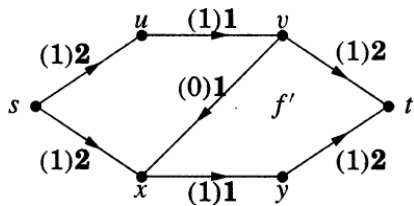
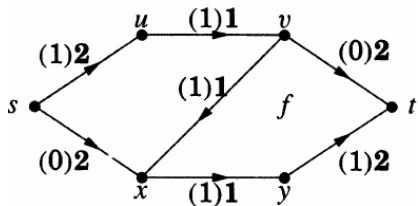
Let P be an f -augmenting path.

$$\epsilon(e) = \begin{cases} c(e) - f(e) & \text{if } e \text{ is forward on } P \\ f(e) & \text{if } e \text{ is backward on } P \end{cases}$$



An f -augmenting path leads to a flow with **larger** value.

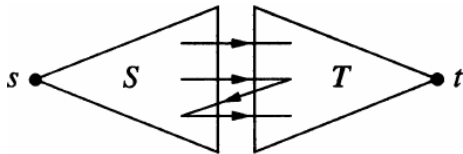
$$\text{tolerance of } P : \min_{e \in E(P)} \epsilon(e)$$

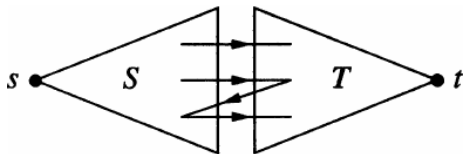


Definition (Source/Sink Cut (割))

In a network, a **source/sink cut** $[S, T]$ consists of the edges from a **source set** S to a **sink set** T , where

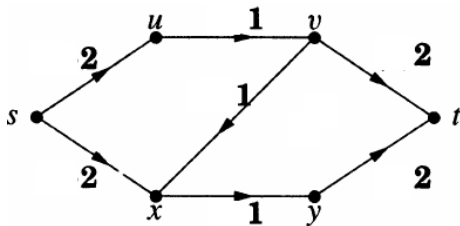
$$(T = V \setminus S) \wedge (s \in S) \wedge (t \in T)$$





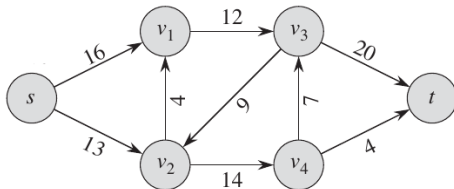
Definition (Capacity of Cut (割的容量))

$$\text{cap}(S, T) = \sum_{u \in S, v \in T, uv \in E} c(u, v)$$



Definition (Minimum Cut (最小割))

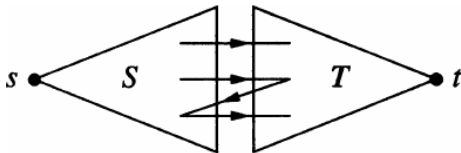
A **minimum cut** is a **cut** of minimum value.

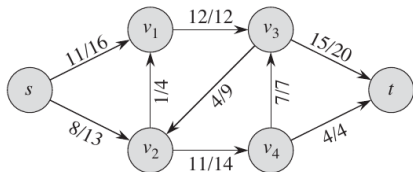
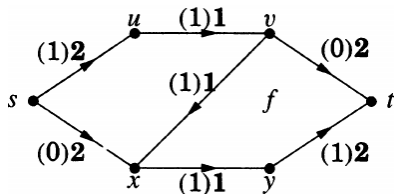


Theorem (Weak Duality (弱对偶定理))

Let f be any feasible *flow* and $[S, T]$ be any source/sink *cut*.

$$\text{val}(f) \leq \text{cap}(S, T).$$

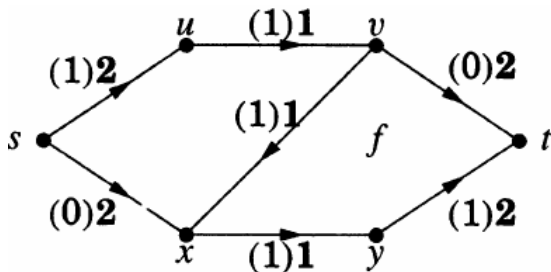




$$f^+(U) = \sum_{u \in U, v \in \bar{U}, (u,v) \in E} f(u, v)$$

$$f^-(U) = \sum_{v \in \bar{U}, u \in U, (v,u) \in E} f(v, u)$$

net flow out of U : $f^+(U) - f^-(U) = \sum_{u \in U} (f^+(u) - f^-(u))$



$$\forall U \subseteq (V - \{s, t\}) \quad f^+(U) = f^-(U)$$

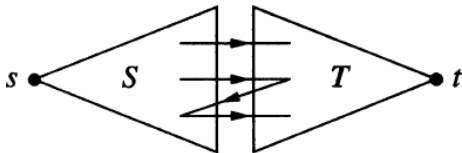
$$s \in U \wedge t \notin U \implies f^+(U) - f^-(U) = f^+(s) = \text{val}(f)$$

$$s \notin U \wedge t \in U \implies f^+(U) - f^-(U) = f^-(t) = \text{val}(f)$$

Theorem (Weak Duality (弱对偶定理))

Let f be any feasible *flow* and $[S, T]$ be any source/sink *cut*.

$$\text{val}(f) \leq \text{cap}(S, T).$$



$$\text{val}(f) = f^+(S) - f^-(S) \leq f^+(S) \leq \text{cap}(S, T)$$

Lemma

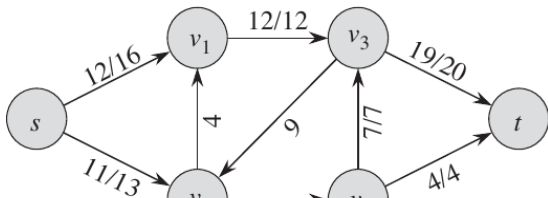
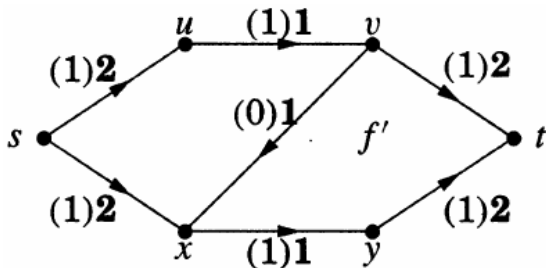
$$\max_f \text{val}(f) \leq \min_{[S,T]} \text{cap}(S, T)$$

What if $\text{val}(f) = \text{cap}(S, T)$ for some flow f and some cut $[S, T]$?

f is maximum and $[S, T]$ is minimum

$$\text{val}(f) = f^+(S) - f^-(S) = f^+(S) = \text{cap}(S, T)$$

$$f^-(S) = 0 \wedge f^+(S) = \text{cap}(S, T)$$



Theorem (Max-flow Min-cut Theorem (Ford and Fulkerson; 1956))

$$\max_f \text{val}(f) = \min_{[S,T]} \text{cap}(S,T)$$

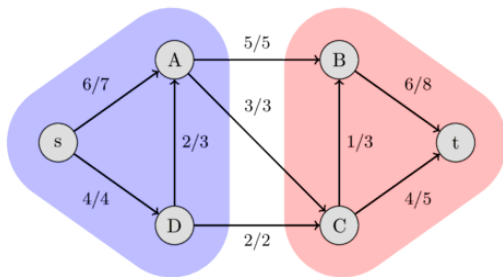
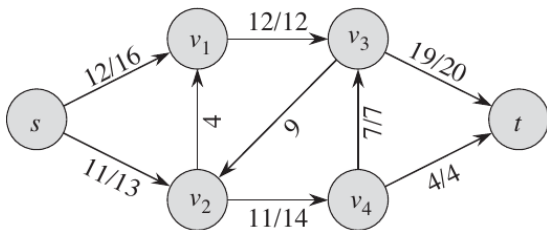
(Strong Duality)



L. R. Ford Jr. (1927 ~ 2017)



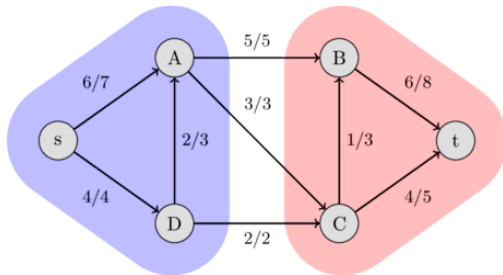
D. R. Fulkerson (1924 ~ 1976)



Theorem

A feasible flow f is maximum iff there are no f -augmenting paths.

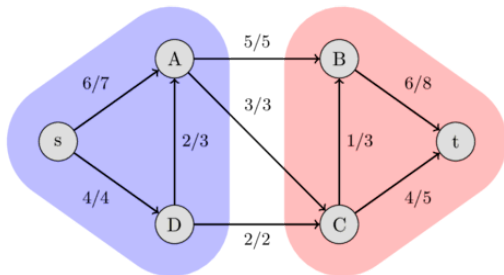
We construct a cut $[S, T]$ with $\text{val}(f) = \text{cap}(S, T)$.



$S \triangleq \{\text{the vertices reachable from } s \text{ along } \textcolor{red}{\text{partial}} \text{ } f\text{-augmenting paths}\}$

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$$T \triangleq V - S$$

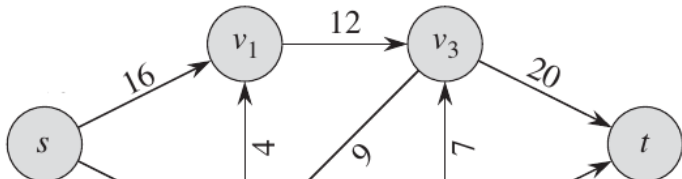
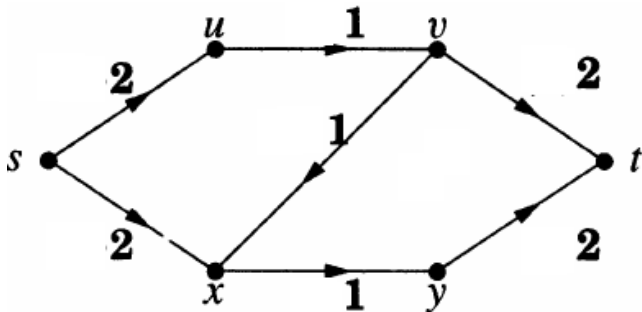


$$f^{-1}(S) = 0 \wedge f^{+}(S) = \text{cap}(S, T)$$

$$\text{val}(f) = f^{+}(S) - f^{-}(S) = f^{+}(S) = \text{cap}(S, T)$$

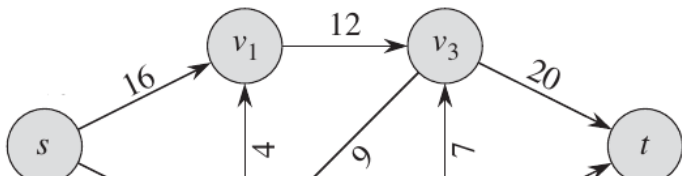
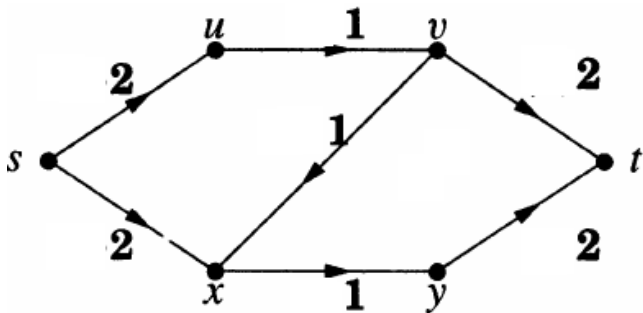
The Ford-Fulkerson Method

Repeatedly **finding** f -augmenting paths until **no more ones exist**.



The Edmonds-Karp Algorithm

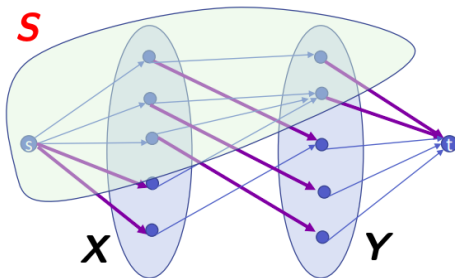
Using **BFS** (Breadth-first Search) to find f -augmenting paths.



Theorem (Hall Theorem; 1935)

There is an *X -perfect matching* of G iff

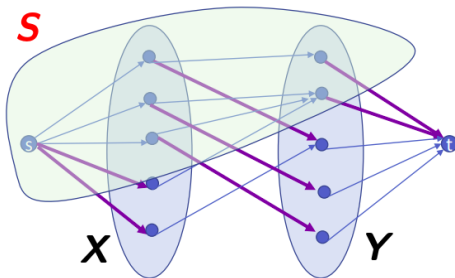
$$\forall W \subseteq X. |W| \leq |N_G(W)|$$



$$\forall x \in X. c(s, x) = 1 \quad \forall y \in Y. c(y, t) = 1 \quad \forall x \in X, y \in Y. c(x, y) = \infty$$

We need to show that $\max_f \text{val}(f) = |X|$.

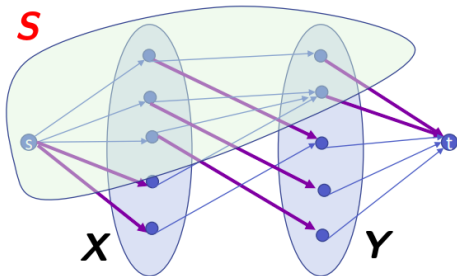
We need to show that $\min_{[S, \bar{S}]} \text{cap}(S, \bar{S}) = |X|$.



$$\min_{[S, \bar{S}]} \text{cap}(S, \bar{S}) \leq |X|$$

Therefore, we need to show that $\min_{[S, \bar{S}]} \text{cap}(S, \bar{S}) \geq |X|$.

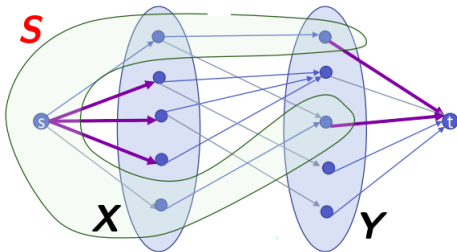
Let $[S, \bar{S}]$ be a cut. We need to show that $\text{cap}(S, \bar{S}) \geq |X|$.



$$N(S \cap X) \not\subseteq (S \cap Y)$$

$$\text{cap}(S, \bar{S}) = \infty \geq |X|$$

$$N(S \cap X) \subseteq (S \cap Y)$$



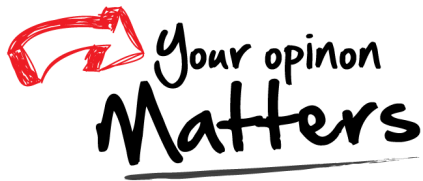
$$\begin{aligned}
 \text{cap}(S, \bar{S}) &= \sum_{u \in S, v \in \bar{S}} c(x, y) \\
 &= \sum_{v \in \bar{S} \cap X} c(\textcolor{red}{s}, v) + \sum_{u \in S \cap Y} c(u, \textcolor{red}{t}) \\
 &= |X| - |S \cap X| + |S \cap Y| \\
 &\geq |X| - |S \cap X| + \textcolor{red}{|N(S \cap X)|} \geq |X|
 \end{aligned}$$

Theorem (König (1931), Egerváry (1931))

If G is a bipartite graph, then the maximum size of a matching in G equals the minimum size of a vertex cover of G



Thank
You!



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